

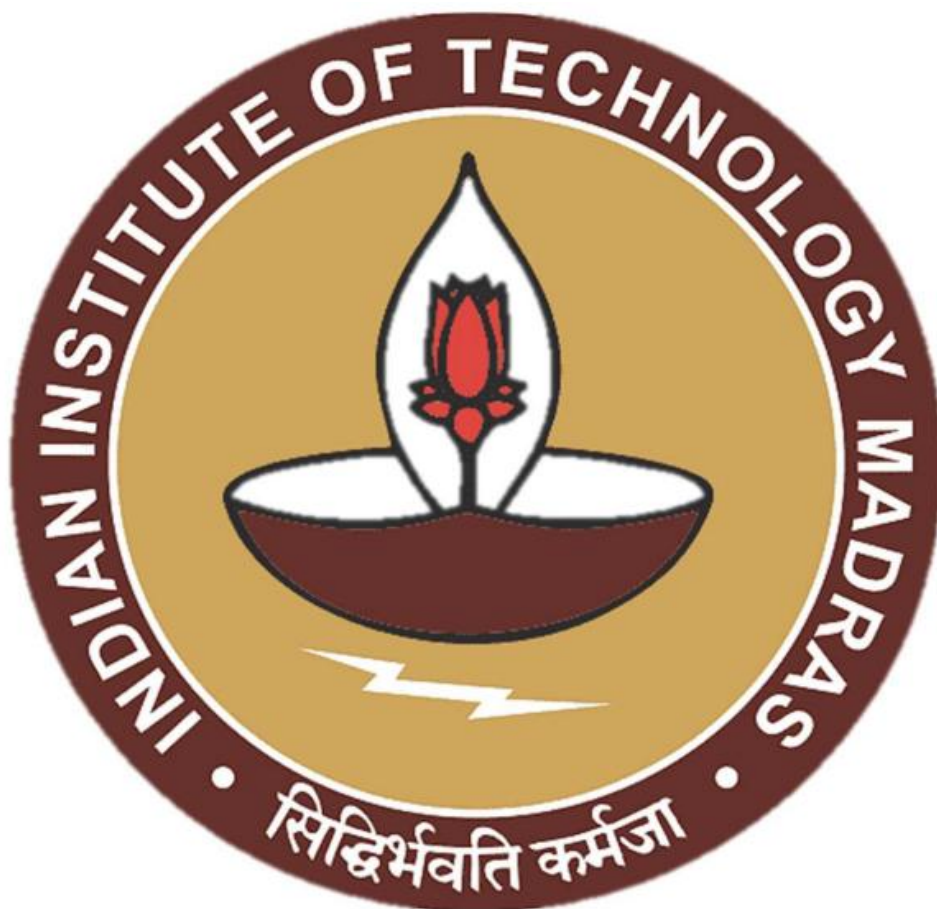
Optimizing Small F&B businesses with Data Analytics

A Mid-Term Report for BDM Capstone Project

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1. Executive Summary

This project, titled “*Optimizing Small F&B Businesses with Data Analytics*”, focuses on *Shubham Sweet Corner*, a small food and beverage shop located at Lakshminiya Chowk, Polytechnic Bypass Road, Saharsa, Bihar. Despite its popularity for snacks and sweets, the business faces three major challenges: lack of proper record-keeping, frequent employee turnover, and limited customer reach due to its distant location from the college area. These issues directly impact profitability, decision-making, and growth potential.

To address these challenges, the project proposes *feasible low-cost and manual solutions* that can be realistically adopted by the shop. First, a *structured manual record-keeping system* in the form of logbooks will be implemented to track daily sales, expenses, and profits. This will replace guesswork with reliable financial insights, enabling the owner to monitor costs and set realistic goals. Second, *employee retention measures* will be analysed, including introducing a refundable security deposit to reduce disruptions in daily operations. Third, *customer demand tracking* will be carried out through daily SKU-level sales logs, helping identify bestselling products and seasonal preferences. Additionally, *location-based marketing strategies* will be explored, such as deploying a temporary pushcart near the college during recess hours to reach more students without requiring high capital investment. Finally, a *cost-benefit analysis* will evaluate the feasibility of future expansion or relocation.

By integrating these structured approaches with simple data visualization, the shop will gain clarity on its financial performance, improve workforce stability, and expand its customer base. The expected outcome is a more efficient, profitable, and sustainable small business that can make informed decisions despite financial constraints.

2. Proof of Originality

Details:

- **Shop Name:** Shubham Sweet Corner
- **Type of Organization:** B2C (Business to Customer)
- **Shop’s Address:** Lakshminiya Chowk, Polytechnic Bypass Road, Saharsa, Bihar.
- **Owner’s Name:** Chandra Kishore Jha

- **Co-Owner's Name:** Shubham Jha
- **No. of Employees:** 3
- **Shop Working Days:** 7 days a week.
- **Shop Working Hours:** 7:00 AM to 9:00 PM



Shop Picture



Shop Owner: Mr. Chandra Kishore Jha

Proof Links:

- **Interaction Video:** [Interaction with the Co-Owner/Manager](#)
- **Video of the Shop:** [Shop Video](#)
- **Inside of the Shop:** [Inside View](#)
- **Picture of the Owner with Co-Owner:** [Photo](#)
- **Authorization Letter:** [Authorization from Shubham Sweet Corner](#)

3. Metadata

In the workbook there are 4 Sheets namely, **Product Pricing**, **Daily Sales Log**, **Employee Details**, and **Variable Expenses**.

Product Pricing	Daily Sales Log	Employee Details	Variable Expenses
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- **Product Pricing:** This Sheet holds the data of all the items sold in the shop, with their cost price, selling price and profit per unit.
- **Daily Sales Log:** This Sheet holds the data of units sold on each day and the Total Revenue generated at the end of the day.
- **Employee Details:** This Sheet holds the data of the names and expenses of employee working in the shop.
- **Variable Expenses:** This Sheet holds the data for expenses like gas and electricity.

a) **Product Pricing Worksheet:**

Product Name	Category	Cost Price (₹)	Selling Price (₹)	Profit per Unit (₹)	Profit Margin (%)
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- **Product Name:** This column represents the name of different products. There are a total of 18 different products listed, namely, *Samosa, Litti, Rasgulla, Gulab Jamun, Laddoo, Ashirwad Milk 1 Kg, Ashirwad Milk ½ Kg, Amul Milk 1 Kg, Amul Milk ½ Kg, Ashirwad Lassi, Dahi 1 Kg, Dahi 200g Pouch, Dahi 400g Pouch, Dahi 200g Dabba, Dahi 400g Dabba, Paneer, Water_Bottle ½ Litre, Water Bottle 1 Litre.*
- **Category:** This was an empty column which was removed during Data Cleaning.
- **Cost Price:** This Column represents the cost price per unit of each SKU. Note, this cost price is calculated with all the variable expenses included.
- **Selling Price:** This Column represents the selling price per unit of each SKU.
- **Profit per Unit:** This Column represents the profit per unit per product, which is calculate by selling price – cost price. The profit here is the net profit after all deductions.

- **Profit Margin:** This column represents the profit margin per unit per product.

It is calculated by the formula, $\frac{\text{profit per unit}}{\text{selling price}} * 100$.

b) Daily Sales Log Worksheet:

Day	Date	Samosa_Qty	Samosa_Selling_Price	Samosa_Total_Revenue	Litti_Qty	Litti_Selling_Price	Litti_Total_Revenue	Rasgulla_Qty	Rasgulla_Selling_Price	Rasgulla_Total_Revenue	Gulab_Jamun_Qty
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- **Day:** This Column represents the day of week for each day.
- **Date:** This Column represents the date when the dataset was collected. The starting date is **15-06-2025** and the ending date is **30-09-2025**. **Note**, there is **no data collected in between 20-07-2025 to 09-08-2025**, as the shop was closed.
- **Samosa_Qty:** This Column represents the quantity of samosa's sold per day. **Note**, similarly the data is collected for each product as mentioned in **Product Name** column.
- **Samosa_Selling_Price:** This Column represents the selling price of each unit of samosas. **Note**, similarly the data is collected for each product as mentioned in **Product Name** column.
- **Samosa_Total_Revenue:** This Column represents the total revenue, which is calculated by, $Qty * Selling Price$. **Note**, similarly the data is collected for each product as mentioned in **Product Name** column.
- **Total_Revenue_per_Day:** This Column represents the total revenue per day across all products.

c) Employee Details Worksheet:

Employee Name	Role	Monthly Salary (₹)	Working Days	Daily Cost (₹)
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- **Employee Name:** This Column represents the name of the employee.
- **Role:** This Column represents the role of the employee.
- **Monthly Salary:** This Column represents the monthly salary of each employee.
- **Working Days:** This Column represents the number of working days in a month, of the employee.
- **Daily Cost:** This Column represents the daily cost of the employee, calculated by, $\frac{\text{Monthly Salary}}{\text{Working Days}}$

d) Variable Expenses Worksheet:

Expense Type	Amount (₹)
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- **Expense Type:** This Column represents the type of expenses from gas, rent or electricity.
- **Amount:** This Column represents amount per expense type.

❖ **Link to the Workbook:** [Food Business Data Collection](#)

4. Descriptive Statistics:

The *Overall Daily Revenue Statistics* of the business is calculated using `.describe()` method in python. In Fig. 4.1, the **count** of **87** represents the *no. of operational business days* in the timeframe the data was collected. The *Average Daily Revenue (mean)* is **₹10333.77**, the *Standard Deviation (std)* is **₹556.67**, the *median* is **₹10397.00**. The *maximum and minimum revenue* is **₹13432.00 and ₹9009.00** respectively. The *Lowest Revenue Day* is on **16-08-2025**, as the top selling SKU, i.e., the Samosa's were not sold that day.

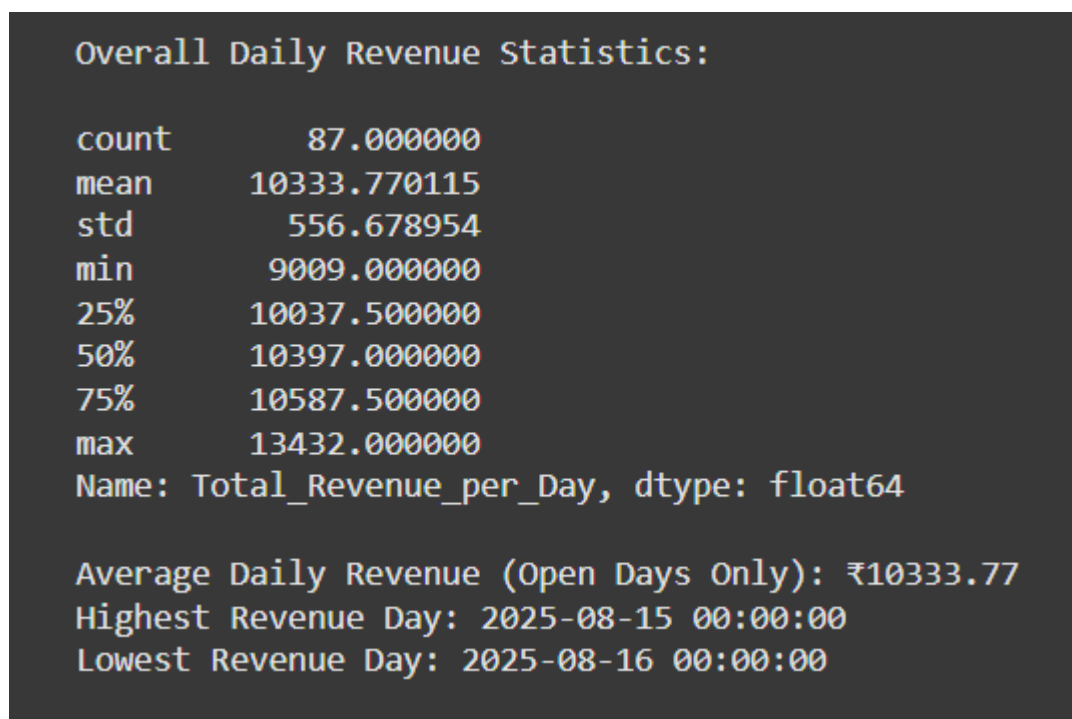


Fig. 4.1 – Representing the overall daily revenue statistics.

Fig. 4.2, represents the product wise summary of the dataset. Where *Avg. Quantity Sold*, is the average number of units sold per day in the timeframe the data was collected. The *Total Revenue*, is the overall revenue generated per product during the 87 observed working days. The *Avg. Daily Revenue*, is the average revenue generated per product.

index	Avg Quantity Sold	Total Revenue (₹)	Avg Daily Revenue (₹)
Samosa	408.13	355070.0	4081.26
Litti	209.03	181860.0	2090.34
Laddoo	4.97	69160.0	794.94
Rasgulla	51.18	66795.0	767.76
Paneer	1.09	28500.0	327.59
Amul_Milk_1_Kg	5.37	27086.0	311.33
Ashirwad_Milk_1_Kg	5.05	24584.0	282.57
Dahi_1_Kg	1.95	19550.0	224.71
Amul_Milk_1/2_Kg	5.06	13200.0	151.72
Ashirwad_Milk_1/2_Kg	5.13	12934.0	148.67
Dahi_400g_Dabba	2.01	8750.0	100.57
Dahi_400g_Pouch	2.23	6790.0	78.05
Water_Bottle_1/2_Litre	7.03	6120.0	70.34
Dahi_200g_Dabba	2.03	5310.0	61.03
Ashirwad_Lassi	4.74	4944.0	56.83
Dahi_200g_Pouch	1.94	3380.0	38.85

Fig. 4.2 – Product Wise Summary of the dataset.

Fig. 4.3, represents the *top 5 products* sold with respect to the total revenue and average revenue generated per day. The *top selling product* is *Samosa* with the *Avg. Quantity Sold* as **408**. Although, it has to be noted that the **profit margin is only 6% for samosas**.

Top 5 Products by Total Revenue:			
	Avg Quantity Sold	Total Revenue (₹)	Avg Daily Revenue (₹)
Samosa	408.13	355070.0	4081.26
Litti	209.03	181860.0	2090.34
Laddoo	4.97	69160.0	794.94
Rasgulla	51.18	66795.0	767.76
Paneer	1.09	28500.0	327.59

Fig. 4.3 – Top 5 Products by Total Revenue

Between **15 June 2025 and 30 September 2025**, Shubham Sweet Corner recorded detailed sales, expense, and employee data to evaluate business performance using descriptive analytics. During this period, the shop remained closed from **20 July to 9 August 2025** due to unforeseen circumstances. These non-operational days were treated as missing values (NaN) to prevent bias in the analysis and ensure that the statistics reflect only active business days.

5. Detailed Explanation of Analysis Process/Method:

The analysis was conducted in a structured, step-by-step manner using *Python and Microsoft Excel* to derive meaningful insights from the manually collected business data of ***Shubham Sweet Corner***. The primary goal was to convert unorganized records into a structured dataset, apply descriptive statistics, and identify key business patterns across sales metrics.

a) Data Collection and Organization

Data was collected manually from the shop on a daily basis for the period **15 June 2025 to 30 September 2025**. Four datasets were prepared in Excel, namely, *Product Pricing*, *Daily Sales Log*, *Employee Details*, and *Variable Expenses*. Each of which has been explained in the metadata part earlier.

Each dataset was cleaned, formatted, and validated before analysis.

b) Data Cleaning and Pre-processing

Before performing statistical analysis, data was cleaned using *Python's pandas* library:

- *Handled missing or inconsistent values* by replacing blank cells with **NaN** where data was not applicable (e.g., shop closure period, Category column).
- Converted all *Date columns* into *datetime* objects for accurate chronological sorting and filtering.
- Ensured all numerical columns were correctly formatted as numeric data types for statistical computation.

A special condition was added to mark the **closure period (20 July 2025 – 9 August 2025)**. During this interval, the shop was non-operational, and the corresponding revenue values were replaced with **NaN** to prevent biased results. This maintained chronological integrity while ensuring analytical accuracy.

c) Descriptive Statistics and Aggregation

Descriptive statistical measures such as *mean*, *median*, *minimum*, *maximum*, and *standard deviation* were computed using *DataFrame.describe()*.

- For the *Product Pricing* dataset, these metrics summarized average profit margins and per-unit profitability.
- In the *Daily Sales Log*, total daily revenue was aggregated from all product revenue columns using Excel, then passing it to python for further analysis.

6. Results and Findings

The descriptive statistical analysis of *Shubham Sweet Corner* from **15 June 2025 to 30 September 2025** yielded several meaningful insights into its financial performance, and sales trends.

=== BUSINESS OVERVIEW SUMMARY ===		
	Metric	Value
0	Average Daily Revenue (₹)	10333.770115
1	Highest Daily Revenue (₹)	13432.000000
2	Lowest Daily Revenue (₹)	9009.000000
3	Average Profit Margin (%)	19.593057
4	Total Variable Expenses (₹)	12000.000000
5	Average Employee Salary (₹)	10566.666667

Fig. 6.1 – The Overview of the Business

- The *Average Profit Margin* is **19.59 %**, and the *Average Daily Revenue* is **₹10333.77**, which makes the *Average daily Profit* to be **₹2024.38**.
- The *top five products* contribute to over **78%** of the revenue. Out of which *Samosa* is the prime SKU with **39.5%** contribution to the revenue, as shown in Fig. 6.2.
- Although, *Samosa*'s are the prime SKU's but the profit margin is only **6%**. With *profit per unit* to be **₹0.6**, as shown is Fig. 6.3.

- The *Highest Daily Revenue* recorded was on **15-08-2025** of **₹13432.00**. The thing to note here is that no *Samosa*’s are sold on this day, as shown in *Fig. 6.4*.

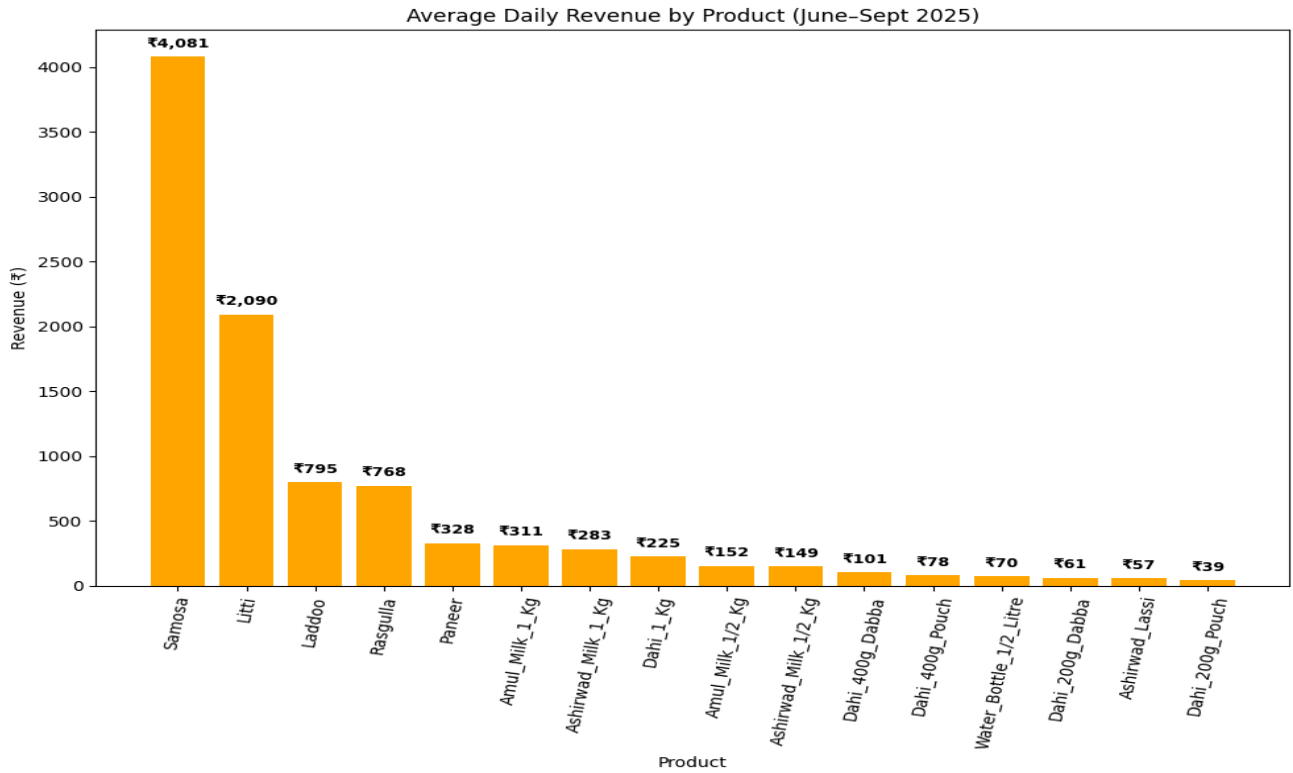


Fig. 6.2- Average daily revenue by Product

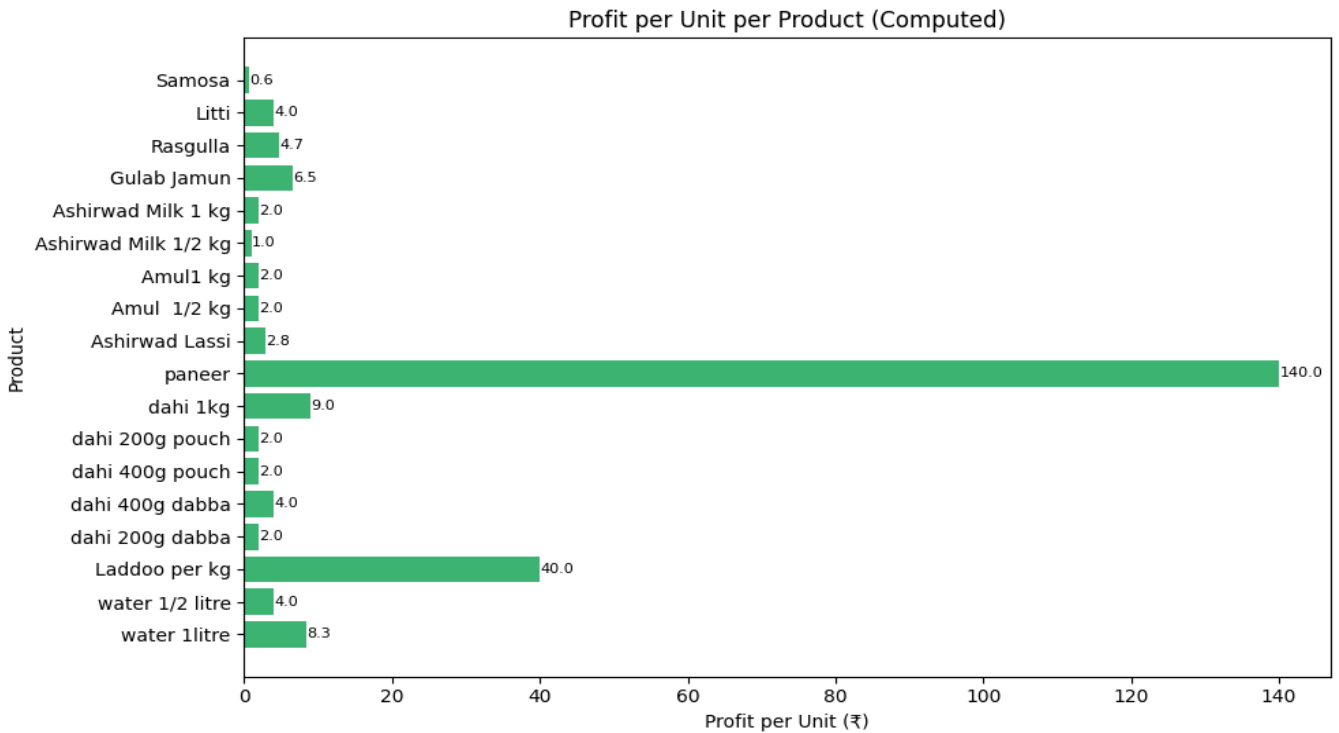


Fig. 6.3- Profit per Unit per Product

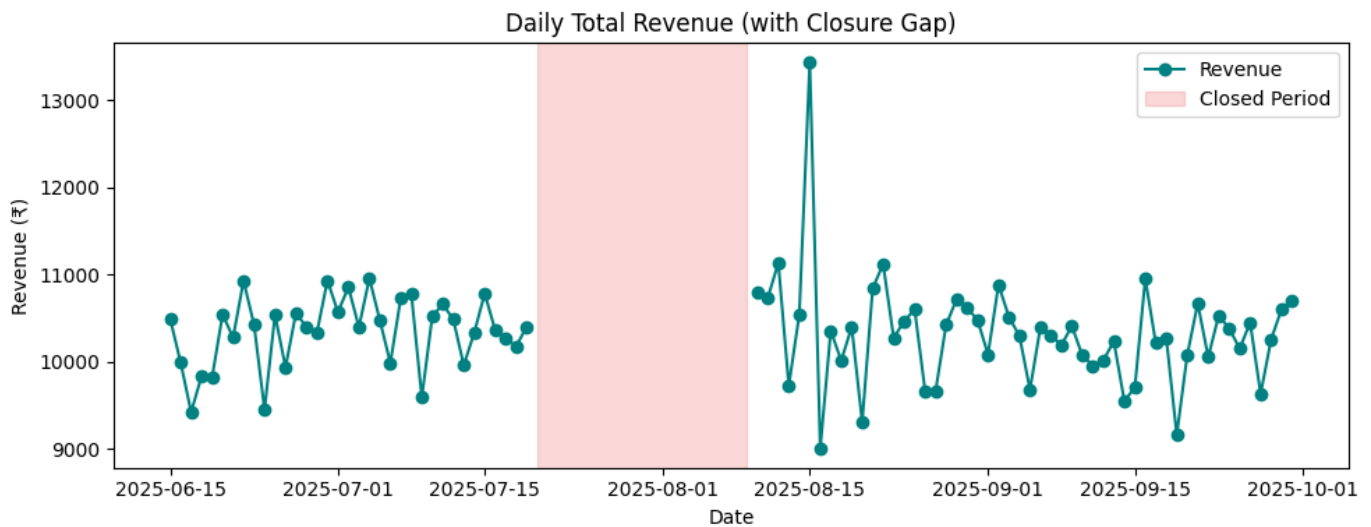


Fig. 6.4- Daily Total Revenue

- The closure period was marked by red, to signify non-operational days, In Fig. 6.4.

Next Steps:

- ❖ *Outliers* if any will be removed to better get insights of the business on normal working days.
- ❖ The arrangement of *the pushcart* will be initiated to increase the sales of Samosa's and Litti's, near the exam centres.
- ❖ *Cost Analysis* will be done for the *prime SKU* as the *profit per unit* is ₹0.6, which is significantly low.
- ❖ A new price will be proposed for the prime SKU after the analysis.
- ❖ Although, already suggested, the *implementation of security deposit* will be done to make sure the employee maintain regularity, as their absence impact the business significantly.
- ❖ *Manual Record Keeping in Excel Workbook* will be initiated for keeping track of every day sales.